

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE

Mid term Exam, Spring 2025

Course Code: CSE 4743

Time: 1 hour and 30 minutes

Course Title: Computer Security

Full Marks: 30

Part A

[Answer the questions from the followings]

1. a) Explain Authenticity and Accountability by showing their use in an Information system. CO1 U 5

Or,

1. a) Explain different types of active attacks. CO1 U 5

1. b) Describe some security mechanisms that are included in ITU X.800 recommendation CO1 U 5

2. a) Explain with example why decrypting one time pad is extremely difficult for an attacker. CO1 U 5

2. b) Analyze the differences between cryptanalysis of Transposition cipher and Caesar cipher. CO2 An 5

Or,

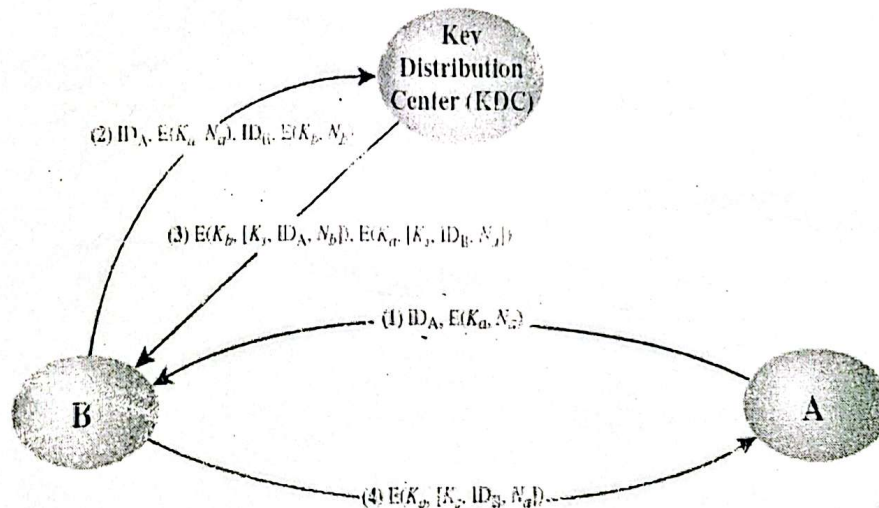
2. b) What is prime field and extension field? Write down the polynomial elements in $GF(2^3)$. CO2 An 5

Part B

[Answer the questions from the followings]

3. a) Why do we need both End to end and Link encryption? Explain with example. CO1 E 5

3. b) Explain each step of the following key distribution scenario. CO2 E 5



International Islamic University Chittagong

Department of Computer Science and Engineering

8th Semester B.Sc. Engineering Mid-term Examination, Spring 2025

Course Code: CSE-4845

Course Title: Distributed Database

Time: 1.5 Hours

Marks: 30

1. a) Write short notes on the following networking concepts:

i) DHCP ii) DNS iii) ARP iv) TCP

4

OR

Explain "A Day in the Life of a Web Page Request", describing the key networking processes involved when a user requests a web page in a browser.

- b) What are distributed databases? Explain the key design considerations for distributed databases.

3

- c) Explain Client-Server and Peer-to-Peer (P2P) architectures of distributed DBMS. How do they differ in terms of structure and communication?

3

2. a) What is fragmentation in a distributed database system? Discuss the role of min-term predicates in fragmentation.

2

- b) Consider two relation schemas *EMP* (*ENo*, *EName*, *Title*) and *PAY* (*Title*, *Salary*). Let $p_1: \text{Salary} < 30000$ and $p_2: \text{Salary} \geq 30000$ be two simple predicates. Perform a horizontal fragmentation of *PAY* with respect to these predicates to obtain *PAY1* and *PAY2*. Using the fragmentation of *PAY*, perform a derived horizontal fragmentation for *EMP*. (You can assume some entries for the relations.)

3

Consider the schema *ORDER* (*ID*, *Product*, *Category*, *Price*, *Qty*, *Status*) and the data shown in Figure 1. Identify the type(s) of fragmentation represented in the figure. Verify the correctness rules for fragmentation. Explain whether each rule is satisfied for the given data and justify your answer.

5

ID	Product	Category	Price	Qty	Status
1001	iPhone 14	Electronics	799	1	Shipped
1002	Samsung Galaxy S22	Electronics	699	1	Delivered
1003	Adidas Sneakers	Fashion	120	2	Processing
1004	Sony Headphones	Electronics	199	1	Shipped
1005	Levi's Jeans	Fashion	55	3	Delivered
1006	Dell Laptop	Electronics	999	1	Processing

ID	Product	Category
1001	iPhone 14	Electronics
1004	Sony Headphones	Electronics

Site 1

ID	Price	Qty	Status
1001	799	1	Shipped
1004	199	1	Shipped

Site 2

ID	Product	Category	Price	Status
1002	Samsung Galaxy S22	Electronics	699	Delivered
1005	Levi's Jeans	Fashion	55	Delivered

ID	Product	Category	Price
1003	Adidas Sneakers	Fashion	120
1006	Dell Laptop	Electronics	999

Site 4

ID	Qty	Status
1003	2	Processing
1006	1	Processing

Site 5

Figure 1: Fragmentation of the *Order* relation in an online shopping system

3. a) Compare eager and lazy data storage methods in a distributed system.

2

OR

How does the Global-and-Local-as-View (GLAV) mapping approach address the limitations of both Local-as-View (LAV) and Global-as-View (GAV) approaches?

- b) "When integrating data from multiple heterogeneous sources, various conflicts arise due to differences in how data is structured, named, or represented". Given Schema S1 and Schema S2, detect and describe the conflicts between them.

4

EMPLOYEE (<u>EmpID</u> INT PRIMARY KEY, FullName VARCHAR(50), DOB DATE, Gender CHAR(1), -- M/F Salary FLOAT, -- in USD Address VARCHAR(100), DepartmentID INT) DEPARTMENT (DepartmentID INT PRIMARY KEY, DeptName VARCHAR(50)) Schema S1	STAFF (<u>StaffNumber</u> VARCHAR(10) PRIMARY KEY, Name VARCHAR(60), BirthDate TEXT, Gender INT, -- 0/1 Salary FLOAT, -- in EUR LocationID INT) LOCATION (LocationID INT PRIMARY KEY, LocationAddress VARCHAR(200)) Schema S2
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- c) Consider three sources:

4

- Database 1 has one relation *Area(Id, Field)* providing areas of specialization of employees; the Id field identifies an employee.
- Database 2 has two relations, *Teach(Professor, Course)* and *In(Course, Field)*; Teach indicates the courses that each professor teaches and In that specifies possible fields that a course can belong to.
- Database 3 has two relations, *Grant(Researcher, GrantNo)* for grants given to researchers, and *For(GrantNo, Field)* indicating which fields the grants are for.

The objective is to build a GCS with two relations: *Works(Id, Project)* stating that an employee works for a particular project, and *Area(Project, Field)* associating projects with one or more fields.

- Provide a LAV mapping between Database 1 and the GCS.
- Suppose one extra relation, *Funds(GrantNo, Project)*, is added to Database 3. Provide a CAV mapping in this case.

International Islamic University Chittagong

Dept. of Computer Science & Engineering (CSE)

B.Sc. in CSE, Mid-Term Examination, Spring 2025

Course Code: CSE 4875 Title: Pattern Recognition and image processing

Total Marks: 50 Time: 1.5 hours

			CO	DL																
1(a)	Define image processing. Write the application areas of image processing.	2✓	CO1	C2																
1(b)	Describe <i>grayscale</i> image and <i>color</i> image. Give necessary examples.	2✓	CO1	C2																
1(c)	Explain the fundamental steps of digital image processing.	3	CO1	C2																
1 (d)	In some devices CCD based cameras are used where other use CMOS based cameras" Justify the significance of the statement and write the benefit of using both.	3	CO1	C2																
	Or, How can you differentiate between <i>Monocular</i> and <i>Binocular</i> vision?																			
2 (a)	How <i>contrast stretching</i> enhance the image? Explain in brief.	2	CO1	C2																
2(b)	A 4x4 bits/pixel original image is given by (given with 3 bits/pixel)	5	CO3	C3																
	<table border="1"> <tr><td>6</td><td>0</td><td>1</td><td>7</td></tr> <tr><td>2</td><td>7</td><td>1</td><td>1</td></tr> <tr><td>2</td><td>1</td><td>0</td><td>2</td></tr> <tr><td>2</td><td>2</td><td>1</td><td>6</td></tr> </table> i. Apply histogram equalization to the image by rounding the resulting image pixels to integer. ii. Sketch the histograms of the original image and the histogram-equalized image. iii. "When automatic enhancement is desired, histogram specification is a good approach"- Do you agree? Why?	6	0	1	7	2	7	1	1	2	1	0	2	2	2	1	6			
6	0	1	7																	
2	7	1	1																	
2	1	0	2																	
2	2	1	6																	
2(c)	Describe the mathematical formation of analog image. Analyze the steps involves in digitalize the analog image.	3	CO1	C2																
	Or, If we convolve and correlation an image with the matrix given below, what would be the relation between the original and modified image?																			
	<table border="1"> <tr><td>•0</td><td>•0</td><td>•0</td></tr> <tr><td>•0</td><td>•0</td><td>•1</td></tr> <tr><td>•0</td><td>•0</td><td>•0</td></tr> </table>	•0	•0	•0	•0	•0	•1	•0	•0	•0										
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•0	•0	•0																		
3(a)	What is the difference between Special Domain Processing and Frequency Domain Processing?	3	CO1	C2																
3(b)	"In some aspects, local histogram equalization is more important than global	3	CO2	C2																

For equalization, justify your answer with examples.

A 4 x 4 original image is given with 3 bits/pixel.

4

CO3

2	3	4	0
3	1	5	2
2	7	3	7
2	4	1	3

- Perform Median filter on the image (Use padding)
- Analyze the differences of both images.

Or,

A 4 x 4 original image is given with 3 bits/pixel.

2	3	4	0
3	1	5	2
2	7	3	7
2	4	1	3

- Perform Lowpass filter on the image separately (Use padding)
- Analyze the statement "Lowpass + Highpass = Original image".

International Islamic University Chittagong

Department of Computer Science & Engineering

B.Sc. in CSE Spring-2025, Midterm Exam

Title: Social, Professional & Ethical Issues in Computing

Course Code: CSE-4805, Full Marks: 30 Time: 1 Hours 30 Minutes

Instructions to the Candidate: Answer All Questions

- | | Marks | CLO |
|---|-------|------|
| 1. a) Computers have evolved significantly from large mainframes to modern personal devices, with advancements in hardware and software shaping their development. Early computers relied on punch cards, whereas today's systems use powerful processors and cloud computing. Discuss how have developments in computer hardware and software influenced the evolution of modern computing? | 5 | CLO1 |
| b) How A school is considering integrating AI-powered tools into its teaching methods. These tools would provide personalized learning experiences for students, track progress, and help teachers identify areas where students need additional support. However, some teachers are concerned that AI might replace human interaction and reduce the personal touch in education.

Analyze the potential benefits and challenges of using AI in education. How would you recommend balancing technology with human involvement to enhance the learning experience? | 5 | CLO3 |
| OR | | |
| 1. Discuss the applications and impact of computer on:
i. education,
ii. business | 5+5 | CLO3 |
| 2. a) With the rise of the internet and digital databases, personal information has become easier to collect, store, and share. In the past, privacy was protected by physical barriers, but today online tracking, surveillance, and data breaches pose new challenges.

How has computer technology changed the concept of privacy over time, and what ethical concerns does this raise? | 5 | CLO1 |
| b) A popular video streaming platform collects and analyzes user watch history, search preferences, and interaction data to recommend content and improve user experience. It also shares this data with advertisers to generate revenue.

Evaluate the ethical and business implications of using user data for profit without direct user compensation. Propose a fair approach that balances user privacy and platform sustainability. | 5 | CLO3 |
| 3. a) Briefly discuss the current scenario of e-commerce in Bangladesh. How proper enactment of ICT act can protect consumer and suppliers of e-commerce in Bangladesh? | 5 | CLO4 |
| b) Social media platforms like Facebook, Twitter, and YouTube face the challenging task of moderating content to prevent offensive and hate speech. While these platforms aim to protect users, their moderation practices often raise ethical concerns regarding freedom of expression.

How can these platforms balance preventing harm with respecting freedom of expression? Discuss the consequences of excessive censorship versus insufficient moderation. | 5 | CLO2 |